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Course Code: **23MAT214**
Mathematics for Intelligent System-4

Matrix Splitting and Proximal Algorithms

Problem Definition:

We want to solve:

$$\min_x \; f(x) + g(x)$$

where
$f(x)$: smooth (differentiable)
$g(x)$: possibly non-smooth

Example: $g(x) = | x |$

👉 Not differentiable at $x = 0$

So:

- Gradient descent ❌ fails
- Classical calculus ❌ fails

The difficulty is that **gradient descent** works only when everything is differentiable. For a smooth function, we would do:

$$x^{(k+1)} = x^{(k)} - \alpha \nabla f(x^{(k)})$$

But here we have $g(x)$, which may not have a gradient (for example $g(x) = | x |$). So we need a way to include $g(x)$ **without differentiating it**.

> ***The idea is: instead of minimizing the full function directly, we solve a simpler approximation around the current point $x^{(k)}$.***

We approximate only the smooth part $f(x)$using first-order Taylor expansion around $x^{(k)}$:

$$f(x) \approx f(x^{(k)}) + \nabla f(x^{(k)})^T (x - x^{(k)})$$

Now if we try to minimize this approximation plus $g(x)$, we get:

$$\min_x \; f(x^{(k)}) + \nabla f(x^{(k)})^T (x - x^{(k)}) + g(x)$$

But this expression is **linear in** x (because Taylor expansion is linear). A linear function has no minimum — it keeps decreasing in some direction and goes to infinity. So, this problem is not solvable.

To fix this, we add a quadratic term:

$$\frac{1}{2\alpha} \| x - x^{(k)} \|^2$$

This is not arbitrary. We add it because:

- it prevents x from moving too far away from $x^{(k)}$
- it makes the problem strongly convex (so a unique minimum exists)
- it converts the problem into something solvable

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Now the new problem becomes:

$$\min_x [f(x^{(k)}) + \nabla f(x^{(k)})^T (x - x^{(k)}) + g(x) + \frac{1}{2\alpha} \| x - x^{(k)} \|^2]$$

Now we simplify this step by step.

First, notice that $f(x^{(k)})$does not depend on x. It is just a constant. Removing a constant does not change where the minimum occurs. So, we drop it:

$$\min_x [\nabla f(x^{(k)})^T (x - x^{(k)}) + g(x) + \frac{1}{2\alpha} \| x - x^{(k)} \|^2]$$

Now expand the first term:

$$\nabla f(x^{(k)})^T (x - x^{(k)}) = \nabla f(x^{(k)})^T x - \nabla f(x^{(k)})^T x^{(k)}$$

Again, the second term is constant (no x), so we drop it.

Now the problem becomes:

$$\min_x [g(x) + \nabla f(x^{(k)})^T x + \frac{1}{2\alpha} \| x - x^{(k)} \|^2]$$

Expand the norm

$$\| x - x^{(k)} \|^2 = (x - x^{(k)})^T (x - x^{(k)})$$

Now multiply:

$$= x^T x - x^T x^{(k)} - (x^{(k)})^T x + (x^{(k)})^T x^{(k)}$$

Use symmetry, we know:

$$x^T x^{(k)} = (x^{(k)})^T x$$

So:

$$\| x - x^{(k)} \|^2 = x^T x - 2(x^{(k)})^T x + (x^{(k)})^T x^{(k)}$$

Multiply by $\frac{1}{2\alpha}$

$$\frac{1}{2\alpha} \| x - x^{(k)} \|^2 = \frac{1}{2\alpha} x^T x - \frac{1}{\alpha} (x^{(k)})^T x + \text{constant}$$

(we ignore the last term since it does not depend on x)

Now the objective becomes:

$$\min_x [g(x) + \nabla f(x^{(k)})^T x + \frac{1}{2\alpha} x^T x - \frac{1}{\alpha} (x^{(k)})^T x]$$

$$\min_x [g(x) + \frac{1}{2\alpha} x^T x + \left(\nabla f(x^{(k)}) - \frac{1}{\alpha} x^{(k)}\right)^T x]$$

Now we want to rewrite entire expression as

$$g(x) + \frac{1}{2\alpha} \| x - v \|^2$$

So we expand that:

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$$\frac{1}{2\alpha} \| x - v \|^2 = \frac{1}{2\alpha} x^T x - \frac{1}{\alpha} v^T x + \text{constant}$$

Compare both expressions

We already have:

$$\frac{1}{2\alpha} x^T x + \left(\nabla f(x^{(k)}) - \frac{1}{\alpha} x^{(k)} \right)^T x$$

We want it to match:

$$\frac{1}{2\alpha} x^T x - \frac{1}{\alpha} v^T x$$

Match coefficients

So:

$$-\frac{1}{\alpha} v = \nabla f(x^{(k)}) - \frac{1}{\alpha} x^{(k)}$$

Multiply both sides by $-\alpha$:

$$v = x^{(k)} - \alpha \nabla f(x^{(k)})$$

Now everything matches perfectly, and the problem simplifies to:

$$\min_x \left[g(x) + \frac{1}{2\alpha} \| x - v \|^2 \right]$$

This is called the **proximal operator step**.

So the full algorithm becomes:

First compute:

$$\boldsymbol{v^{(k)} = x^{(k)} - \alpha \nabla f(x^{(k)})}$$

Then compute:

$$\boldsymbol{x^{(k+1)} = \arg\min_x \left[g(x) + \frac{1}{2\alpha} \| x - v^{(k)} \|^2 \right]}$$

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📘 Example 1: Simple 1D L1 Regularization (Soft Thresholding)

We want to solve:

$$\min_x \ \frac{1}{2}(x-5)^2 + |x|$$

Here:

- $f(x) = \frac{1}{2}(x-5)^2$ (smooth)
- $g(x) = |x|$ (non-smooth)

Step 1: Gradient of $f(x)$

$$\nabla f(x) = x - 5$$

Step 2: Proximal Gradient Update Rule

$$v^{(k)} = x^{(k)} - \alpha \nabla f(x^{(k)})$$
$$x^{(k+1)} = \arg\min_x \left[|x| + \frac{1}{2\alpha}(x - v^{(k)})^2 \right]$$

Let step size $\alpha = 1$

Start Iterations

🔵 **Iteration 0 → 1**

Step A: Gradient step

Start with:

$$\boldsymbol{x^{(0)} = 0}$$

Compute gradient:

$$\boldsymbol{\nabla f(0) = 0 - 5 = -5}$$

Now compute:

$$\boldsymbol{v^{(0)} = 0 - (-5) = 5}$$

Step B: Prox step

Now solve:

$$\boldsymbol{\min_x \ |x| + \frac{1}{2}(x-5)^2}$$

Solve this manually (step-by-step)

Case 1: $x > 0$

$$\boldsymbol{F(x) = x + \frac{1}{2}(x-5)^2}$$

Derivative:

$$\boldsymbol{F'(x) = 1 + (x-5) = x - 4}$$

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Set to zero:

$$x = 4$$

✓ **valid (since $x > 0$)**

Case 2: $x < 0$

$$F(x) = -x + \frac{1}{2}(x-5)^2$$
$$F'(x) = -1 + (x-5) = x - 6$$
$$x = 6 \text{ ❌ invalid (not } < 0)$$

Case 3: $x = 0$

$$F(0) = 12.5$$

Compare

- $x = 4 \Rightarrow F = 4.5$
- $x = 0 \Rightarrow F = 12.5$

✅ **Choose minimum**

$$x^{(1)} = 4$$

🔍 **Interpretation**

- Gradient step pulls toward 5
- Prox step reduces magnitude by 1
  👉 Result balances both terms

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📘 **Example 2: Lasso-Type Problem (Feature Selection Idea)**

We want to solve:

$$\min_x \ \frac{1}{2}(2x-4)^2 + 2\,|x|$$

Step 1: Identify functions

$$f(x) = \frac{1}{2}(2x-4)^2, g(x) = 2\,|x|$$

Step 2: Gradient

$$\nabla f(x) = 2(2x-4) = 4x - 8$$

Step 3: Update rule

$$v^{(k)} = x^{(k)} - \alpha(4x^{(k)} - 8)$$

Take $\alpha = 0.25$

Step 4: Iterations

Initial:

$$x^{(0)} = 0$$

Iteration 1

$$v^{(0)} = 0 - 0.25(-8) = 2$$

Prox step:

$$\min_x \; 2 \mid x \mid + \frac{1}{2}(x-2)^2$$

Let's solve it manually (to prove it)

Case 1: $x > 0$

$$F(x) = 2x + \frac{1}{2}(x-2)^2$$
$$F'(x) = 2 + (x-2) = x$$

Set to zero:

$$x = 0$$

But $x > 0$ ❌ invalid

Case 2: $x < 0$

$$F(x) = -2x + \frac{1}{2}(x-2)^2$$
$$F'(x) = -2 + (x-2) = x - 4$$
$$x = 4$$

But $x < 0$ ❌ invalid

Case 3: $x = 0$

$$F(0) = 0 + \frac{1}{2}(4) = 2$$

Final answer:

$$x = 0$$

Iteration 2

$$v^{(1)} = 0 - 0.25(-8) = 2$$
$$x^{(2)} = 0$$

✅ **Final Answer:**

$$x = 0$$

🔍 **Interpretation**

- Data suggests $x = 2$
- Regularization forces $x = 0$
  👉 Feature is removed (sparsity)

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📘 **Example 3: 2D Vector Case (More Realistic ML Example)**

We solve:

$$\min_{x\in\mathbb{R}^2} \frac{1}{2} \| x - b \|^2 + \lambda \| x \|_1$$

Let:

$$b = \begin{bmatrix} 3 \\ -1 \end{bmatrix}, \lambda = 1$$

Step 1: Identify

$$f(x) = \frac{1}{2} \| x - b \|^2, g(x) = \| x \|_1$$

Step 2: Gradient

$$\nabla f(x) = x - b$$

Step 3: Gradient step

$$v = x^{(k)} - \alpha(x^{(k)} - b)$$

Take $\alpha = 1$

$$v = b = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

Step 4: Prox step (component-wise)

We solve:

$$\min_x \| x \|_1 + \frac{1}{2} \| x - v \|^2$$

Apply soft-thresholding to each component:

First component:

$$x_1 = 3 - 1 = 2$$

Second component:

$$x_2 = -1 + 1 = 0$$

✅ **Final Answer:**

$$x = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

🔍 **Interpretation**

- First feature strong → kept
- Second feature weak → removed

👉 This is **feature selection in ML**

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>>>

Differentiate:

$$F'(x) = \lambda + (x - v)$$

Set derivative = 0:

$$\lambda + x - v = 0 \Rightarrow x = v - \lambda$$

Validity condition:

We assumed $x > 0$, so:

$$v - \lambda > 0 \Rightarrow v > \lambda$$

- **Case 2: $x < 0$**

Then:

$$| x | = -x$$

Function becomes:

$$F(x) = -\lambda x + \frac{1}{2}(x - v)^2$$

Differentiate:

$$F'(x) = -\lambda + (x - v)$$

Set derivative = 0:

$$-\lambda + x - v = 0 \Rightarrow x = v + \lambda$$

Validity condition:

We assumed $x < 0$, so:

$$v + \lambda < 0 \Rightarrow v < -\lambda$$

- **Case 3: $x = 0$**

If neither of the above conditions holds, we check $x = 0$.

This happens when:

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$$-\lambda \leq v \leq \lambda$$

✅ **Final Result (Soft-Thresholding)**

$$x = \begin{cases} v - \lambda & v > \lambda \\ 0 & |v| \leq \lambda \\ v + \lambda & v < -\lambda \end{cases}$$

🔥 **Interpretation**

- If $v$is large → reduce it by λ
- If $v$is small → shrink it to 0

👉 This creates **sparsity**

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